Year 12 Maths - Probability

Similar questions - Discrete random variables

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The discrete random variable X has the probability distribution shown, where p is a constant.

x	0	1	2	3
$\mathrm{P}(X=x)$	0.1	p	p^2	0.15

(a) Show that $p^2+p-0.75=0$ and hence find the value of p, explaining why the other root is rejected. **[4]**

(b) For this value of p, find $\mathrm{E}(X)$ and $\mathrm{Var}(X)$. **[4]**

(c) Two independent values of X are chosen. Find the probability that their sum is 3. **[3]**

2 The random variable X takes the values $1,2,3,4$ and no others. Its distribution satisfies $\mathrm{P}(X=1)=a$ and $\mathrm{P}(X=x)=\frac{2}{3}\mathrm{P}(X=x-1)$ for $x=2,3,4$.

(a) Show that $a=\dfrac{27}{65}$. **[3]**

(b) Find the probability that X is even. **[2]**

(c) Two independent values of X are chosen. Find the probability that their total is 5. **[3]**

(d) Find $\mathrm{P}(X \geqslant 3 \mid X \geqslant 2)$. **[2]**

Question 1

(a) Probabilities sum to 1: $0.1+p+p^2+0.15=1$, so $p^2+p-0.75=0$ and $(p-\frac{1}{2})(p+\frac{3}{2})=0$. $p=\frac{1}{2}$; $p=-\frac{3}{2}$ is rejected as a probability cannot be negative

(b) $\mathrm{E}(X)=0(0.1)+1(0.5)+2(0.25)+3(0.15)=1.45$. $\mathrm{E}(X^2)=0.5+4(0.25)+9(0.15)=2.85$. $\mathrm{Var}(X)=2.85-1.45^2=0.7475$

(c) $(0,3),(3,0),(1,2),(2,1)$: $2(0.1)(0.15)+2(0.5)(0.25)=0.03+0.25=0.28$

Question 2

(a) Probabilities are $a, \frac{2}{3}a, \frac{4}{9}a, \frac{8}{27}a$ with sum $\dfrac{27+18+12+8}{27}a=\dfrac{65}{27}a=1$, so $a=\dfrac{27}{65}$

(b) $\mathrm{P}(2)+\mathrm{P}(4)=\dfrac{18}{65}+\dfrac{8}{65}=\dfrac{26}{65}=\dfrac{2}{5}$

(c) $(1,4),(4,1),(2,3),(3,2)$: $2\left(\dfrac{27\times 8}{65^2}+\dfrac{18\times 12}{65^2}\right)=\dfrac{864}{4225}$

(d) $\dfrac{\mathrm{P}(X\geqslant 3)}{\mathrm{P}(X\geqslant 2)}=\dfrac{20/65}{38/65}=\dfrac{10}{19}$